GOSIM

GOSIM Shenzhen 2026

October 16-17, 2026 Shenzhen, China



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GOSIM Shenzhen 2026

State of harness engineering

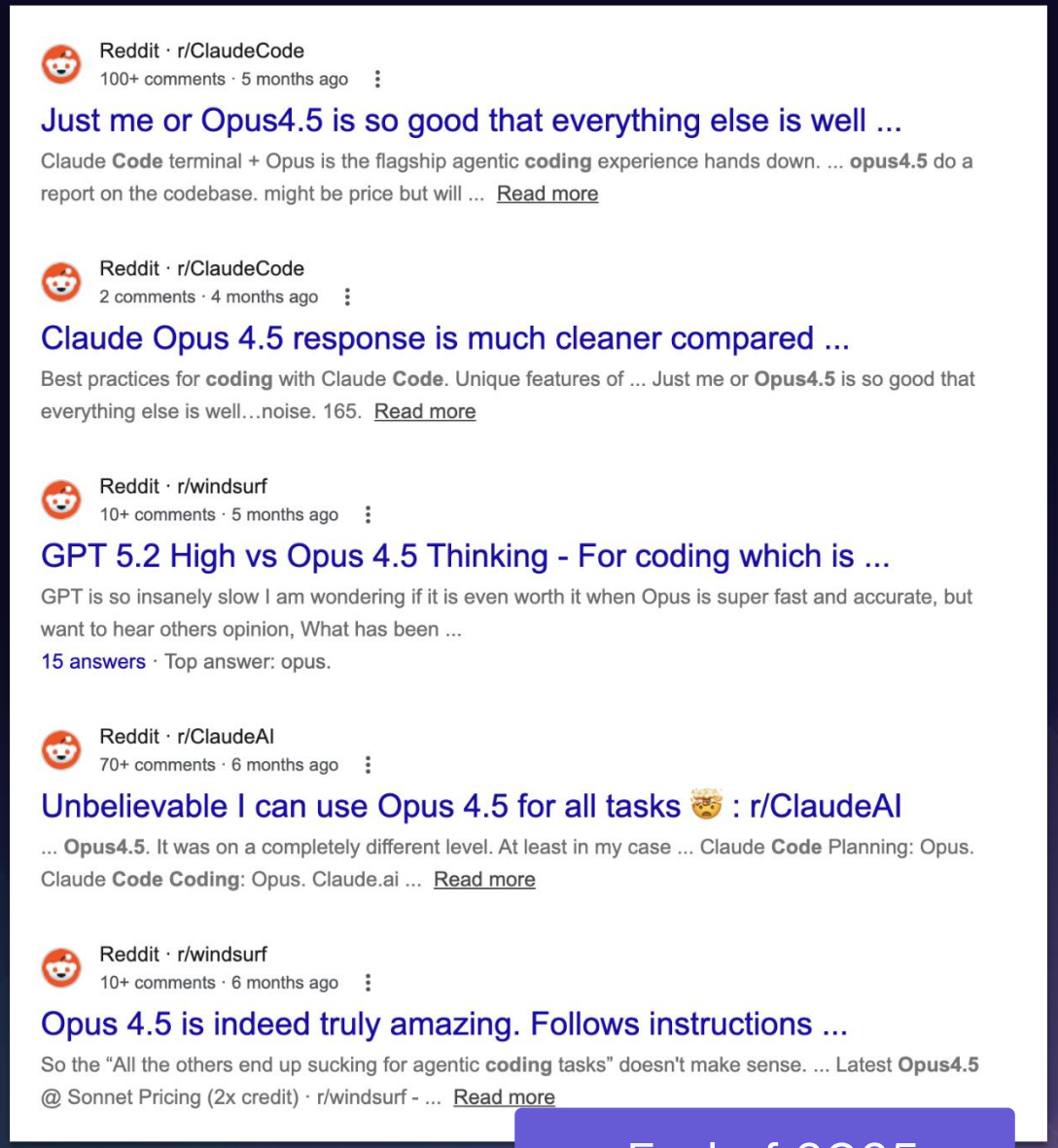
Birgitta Boeckeler
Principal Engineer, Unblocked



From autocomplete to agents - we're four years in

2023	2024	2025	2026
Autocomplete	Claude 3.5 Sonnet Jun	Vibe coding Feb	Swarms Jan
Chats in the IDE May	MCP Nov	Claude Code Feb	OpenClaw Jan
Ask the codebase July	Agentic coding Nov	Background agents May	**Harness engineering Feb**
		Context engineering Jun	Loop engineering Jun
		Skills Oct	Software factories
		Claude Opus 4.5 Nov	

Better models



End of 2025

Better models
...but still lots of supervision

How can we increase our trust in coding agents?

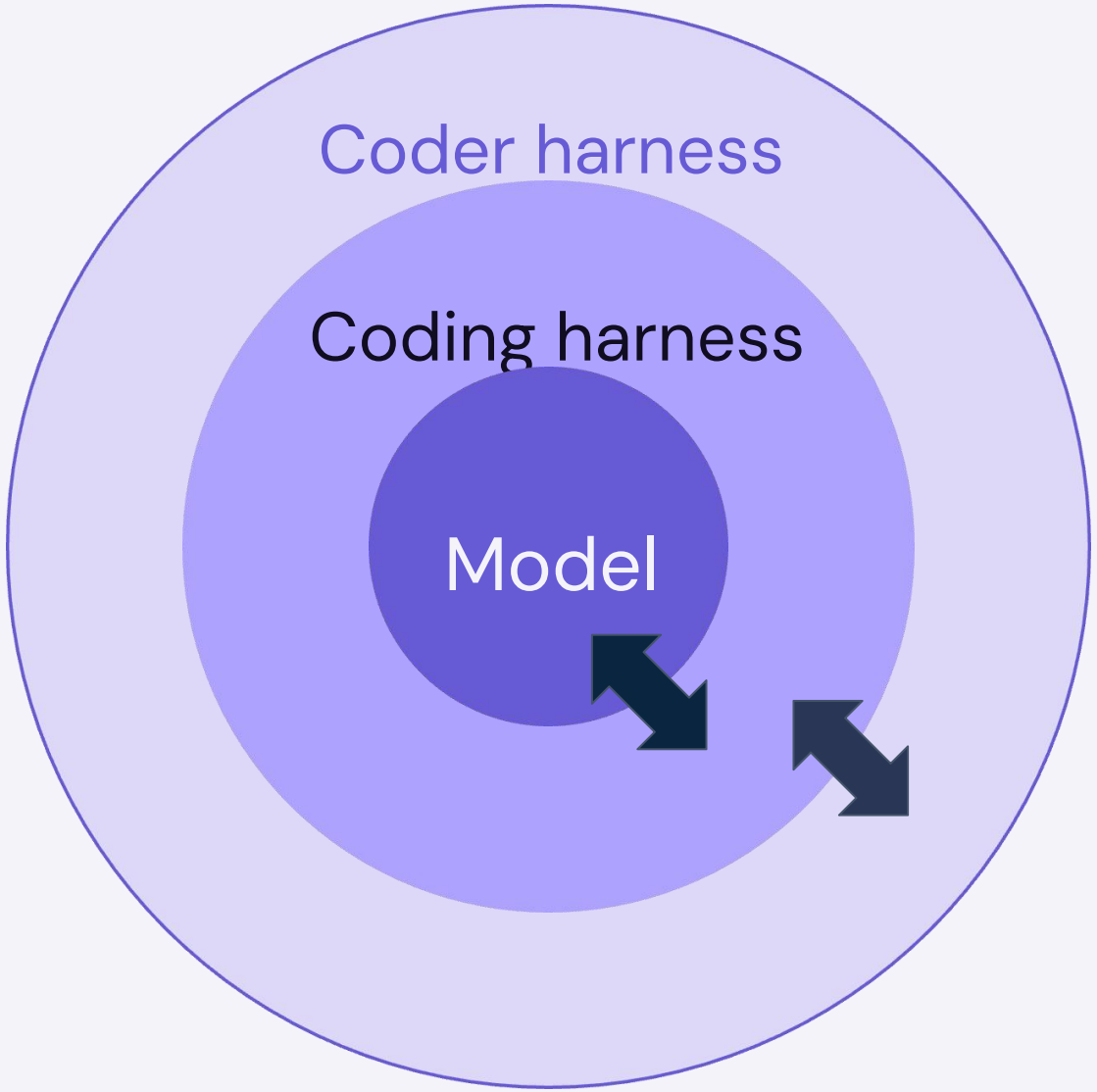
Enter the harness



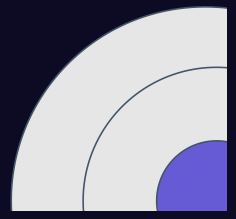
驾驭

缰绳

马鞍



- **Coder harness**
  Skills, AGENTS.md, MCP servers, ...
- **Coding harness**
  Codex, Trae, Deepseek Harness, ...
- **Model**

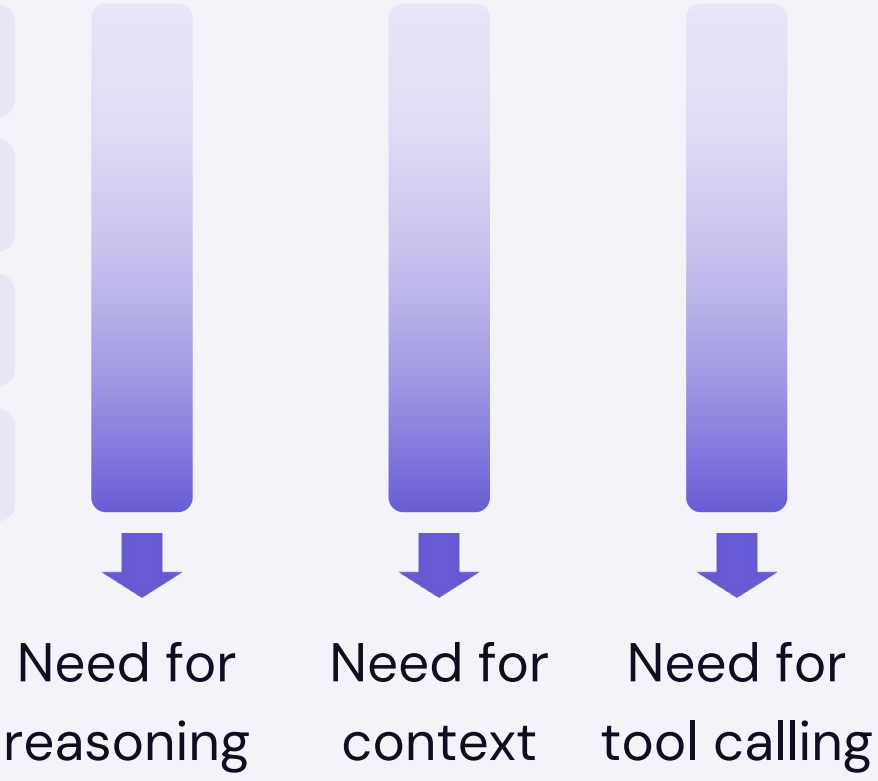


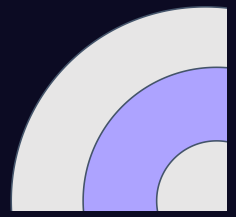
Model

- Not magic, fallible
- Stateless
- Hosting
- Data retention
- Context windows and attention
- Which model for which task

Core skill: **Understanding the task** as it relates to models

- Autocomplete
- Change a few specific files with clear instructions
- Larger, more complex changes with code research
- Planning, debugging, designing





Harness

Understand harness features

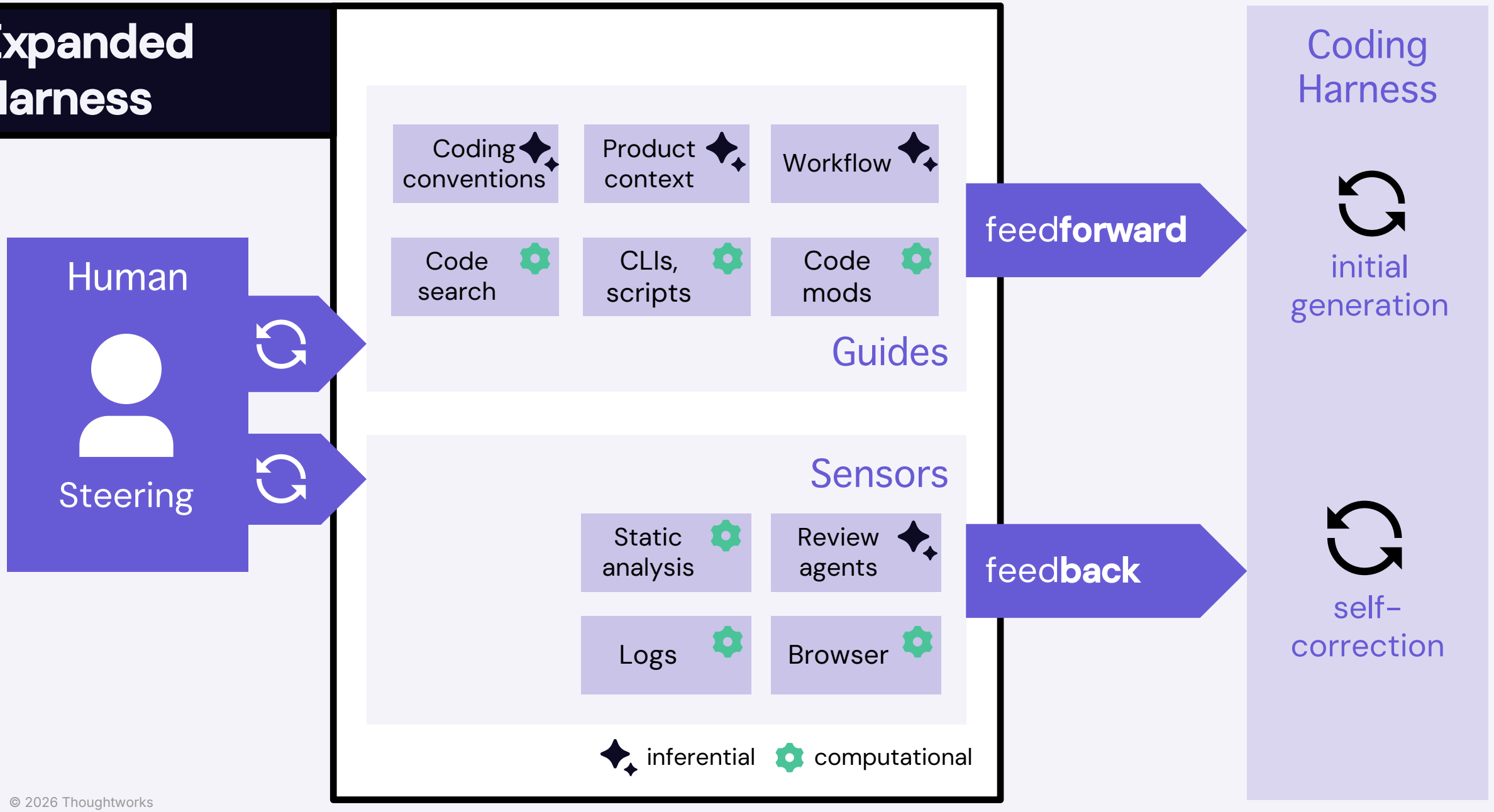
Tool integrations	Orchestration	User interface
Sandboxing	Observability	Ability to run in the cloud
Token footprint	Extensibility	...

The hardest part:

How to use the harness

#context-engineering

Expanded Harness



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Example

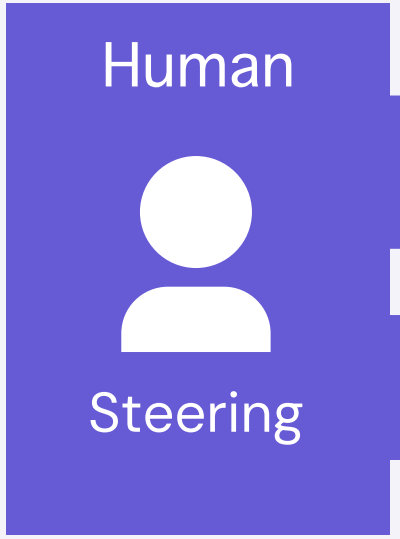
Guide · AGENTS.md

Do not use console.log, use our logger in ./server/logger.ts instead so we always have structured logging.

Sensor · lint rule

no-console: Use `logger` from `server/logger.ts` instead of `console`

Providing context



[Harness engineering] is the idea that anytime you find an agent makes a mistake, you take the time to **engineer a solution such that the agent never makes that mistake again**.

Mitchell Hashimoto, "My AI Adoption Journey"

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Context != Context

Normative

Coding conventions

Architecture rules

Security rules

Last year

AGENTS.md

Instructive

Workflows

Spec-driven flows

How-tos

Last year

AGENTS.md

Informative

Product context

Past decisions

Reference docs

Last year

AGENTS.md

Adding more markdown is not the solution

80%

of Claude Code's system prompt cut for generation 5 models

Anthropic, Jul 2026

< 0

Machine-generated context files did worse than none

ETH Zurich, Feb 2026

4.4%

of CLAUDE.md security rules are enforced

“When ‘do not’ is not deny”, arXiv, Aug 2026

Context != Context

Normative

Coding conventions

Architecture rules

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Today

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MCP Servers

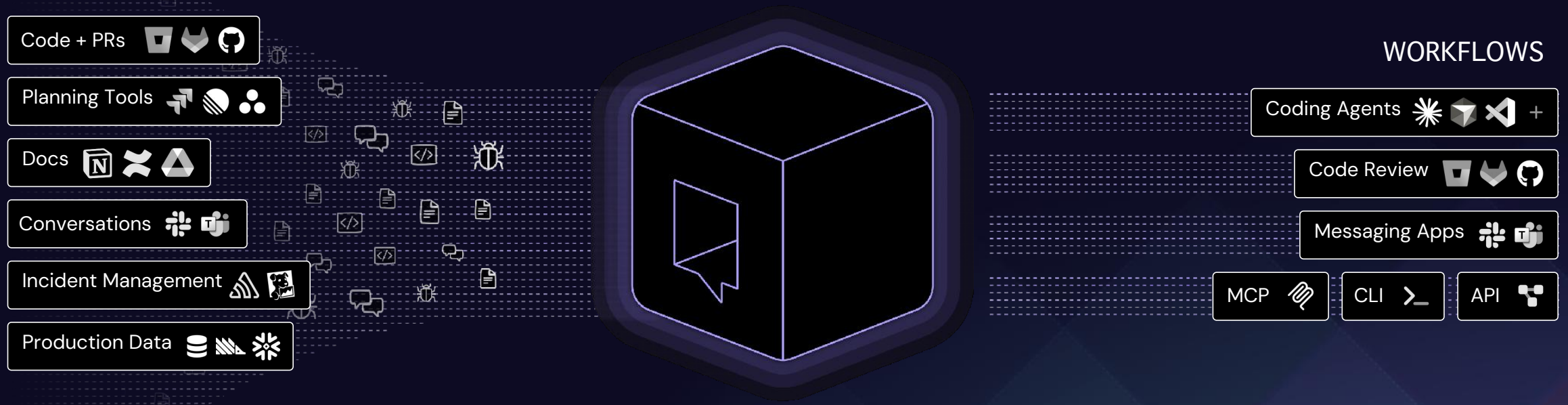
CLIs

Context Engines

Unblocked

How a context engine works

Unblocked



Reasoning	Memory
Resolution	Permissions

How to assess **trust**

Probability



...that the agent gets something wrong

Know the tool and the context
Know your requirements

Impact



...if the agent gets something wrong

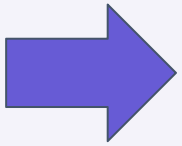
Use case criticality

Detectability



...that the agent got something wrong

Know your feedback loops



Which workflow?

How much review?

How long unsupervised?

Thank you

martinfowler.com/articles/harness-engineering.html

martinfowler.com/articles/sensors-for-coding-agents.html

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